Candiates Name:_____Signature:_____

545/1
**Chemistry**
Paper 1
**July/August 2026**
2hours

# EXCEL ASSESSMENT HUB, WAKISO

## Uganda Certificate of Education

CHEMISTRY

Theory
**Paper 1**
2 hours

**INSTRUCTIONS TO CANDIDATES:**

*This examination paper has* **two** *sections* **A** *and* **B***. it consists of* **six** *examination items.*

*Section* **A** *has* **two** *compulsory items. Responses to section* **A must** *be written in the space provided*

*Section* **B** *has* **two** *parts;* **I** *and* **II***. Respond to only* **one** *item from each part.*

*Responses to section* **B must** *be written in the answer booklet(s) provided*

*Respond to* **four** *examination items in all.*

*Any* **additional** *item(s) answered will* **not** *be scored.*

***For Scorer's Use Only***

| Item No | 1 | | | 2 | | | _____ | | | _____ | | | Total Weighted Score |
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## SECTION A

*Respond to* **all** *items in this section*

**Item 1**

A renewable energy startup is exploring the use of specific alkaline metals to manufacture affordable, rechargeable batteries for off grid solar power systems. The engineering team has received three pure elemental metal samples from a supplier, temporarily labeled **Metal A (Atomic number 3)**, **Metal B (Atomic number 11)**, and **Metal C (Atomic number 19)**.

Before production can begin, the chief engineer needs to categorize these materials correctly to determine the most suitable metal for the battery cells. They must also understand how these metals will behave physically under the varying temperatures of a solar setup, and establish strict safety protocols for the factory workers, as there are known risks of severe fires if these materials are mishandled or exposed to moisture.

You have been hired as the lead chemical consultant to help the engineering team solve this problem.

**Task:**

**a)** Draw the electronic structure for Metals A, B, and C.

**b)** Using their electronic structures, identify the category of the elements in the Periodic Table with a reason.

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**c)** The engineers need to know how these metals will physically behave before using them. Based on the category identified in b) state physical properties that Metals A, B, and C share.

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**d)** During preliminary testing, the engineers observed that Metal C melts much faster than Metal A when exposed to the same amount of heat in the workshop. Explain the trend in the melting points of these metals as you move from Metal A to Metal C, linking it to their atomic structure.

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**e)** Discuss the major safety hazard (side effect) associated with handling and storing these specific metals in the factory, and propose a practical solution the startup must implement to prevent these hazards.

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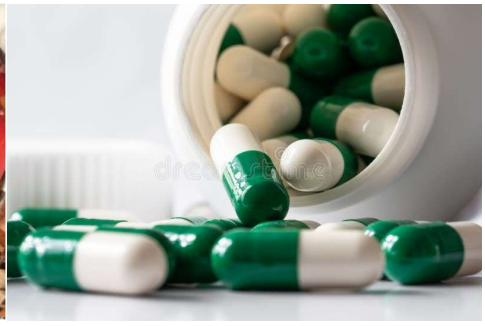
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**Item 2**

A rural sub-county in Nakasongola district has recently experienced a sudden outbreak of bacterial skin and soft tissue infections among livestock farmers. The local community is divided on how to treat the spreading illness.

Some elders are advising the infected youth to use raw extracts crushed from the leaves of a local plant, *Aframomum angustifolium* (wild cardamom), which has been used for generations to heal wounds. Meanwhile, a newly deployed community health officer is urging everyone to visit the health center to be prescribed manufactured **penicillin capsules** to stop the bacteria completely. Because of the conflicting advice, many villagers are using both remedies incorrectly, leading to prolonged sickness or recurring infections.





The local council has organized a sensitization meeting at the community center and has invited you, as a senior chemistry learner, to present a clear write up that guides the community on how these antibacterial options compare, how they work, and how to use them safely.

**Task:**

a) Classify both the medicines used by the village members using your knowledge of chemistry.

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b) Explain how the medicines generally work inside the body.

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c) Compare the two types of medicines by stating
   i) The similarity and,

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   ii) The differences between them

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d) Identify the side effects of using these medicines incorrectly, and give practical solutions to them.

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| Basis Code | T | D | E |
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## SECTION B

## Part I

*Respond to only* **one** *item in this part*

**Item 3**

With the rapid expansion of modern real estate and construction across the country, there is a skyrocketing demand for lightweight, corrosion-resistant materials like aluminium, which are essential for sliding windows, doors, and roofing profiles. Currently, relying heavily on imported aluminium is expensive for local builders.

A visionary investor is planning to set up an industrial extraction plant near a major hydroelectric power station. The plant will process imported bauxite ore into pure aluminium to supply the local construction sector. Before the investor can secure full funding and environmental clearance, they need a detailed technical and impact report to present to the board of directors and the local community leaders.

You have been appointed as the lead chemical engineer to compile this report.

**Task:**

Write a structured report for the investor that addresses the following:

i) the industrial process of extracting aluminium from bauxite (alumina)
ii) ways the establishment of this extraction plant will socially benefit the local community
iii) major environmental side effects directly associated with the aluminium extraction process and propose a solution for each to reassure the community.

**Item 4**

A large sugar manufacturing estate in Eastern Uganda produces thousands of liters of molasses every week as a byproduct of sugar crystallization. Currently, much of this thick, dark syrup is sold cheaply as animal feed or sometimes disposed of inefficiently.

To maximize profits and embrace green energy, the company's management board has decided to set up a new plant to manufacture ethanol from this excess molasses. Their goal is to produce high grade ethanol to be sold as medical disinfectants to local hospitals and as a biofuel additive to be blended with petrol for vehicles. However, the board needs a clear technical breakdown of the production process, social benefits and a risk assessment before they approve the project.

You are the chemical consultant hired to write this proposal for the board.

**Task:**

Write a structured technical proposal that you will present to the board.

## Part II

*Respond to only* **one** *item in this part*

**Item 5**

During hot afternoons, residents and commuters in central Kampala experience stifling heat, heavy smog, and thick exhaust fumes. Environmental scientists have noted that the city's air quality has sharply declined and localized temperatures are rising. This is largely driven by traffic congestion involving older, poorly maintained vehicles.





In response, an environmental NGO has organized a sensitization workshop for local taxi and bus drivers to help them understand how their daily activities impact the atmosphere. You have been invited to the workshop as a chemistry learner to give a talk that helps the drivers understand the situation.

**Task:**

As a learner of chemistry, prepare a write-up that you will present during the workshop.

**Item 6**

The discovery and commercial extraction of crude oil (petroleum) in the Albertine region of Western Uganda has drawn massive investments, including plans for a domestic oil refinery and an export pipeline. The project is expected to process thousands of barrels of crude oil daily into useful products like petrol, diesel, kerosene, and heavy fuel oil to meet the energy demands of the region.

However, local community leaders and environmental civil society groups have expressed deep concerns. They fear that the large-scale exploitation of this underground resource could ruin their environment, accelerate climate change, and disrupt their agricultural livelihoods. To bridge the gap, the petroleum oil company has organized a community town hall meeting and invited you as a chemistry learner to give a clear presentation that balances the benefits against the risks.

**Task:**

As a learner of chemistry, prepare a write-up that you will present during the community town hall meeting.